DBMS LAB 6

Parth Janardan Naukudkar_ECS_II_66_D6

Aim: Understanding and applying INNER JOIN, LEFT OUTER JOIN, RIGHT OUTER JOIN, FULL OUTER JOIN, SELF JOIN, CROSS JOIN, and Multi-table Joins on relational schemas.

1 [Inner Join]: Write an SQL query using INNER JOIN to display the MatchID, Ground, Date, and the Player's full name (FName, LName) for every player who batted in each match.

Solution:

```
SELECT O.MatchID, O.Ground, O.Date, P.FName, P.LName FROM ODIMatch O INNER JOIN Batting B ON O.MatchID = B.MatchID INNER JOIN Player P ON B.PID = P.PlayerID;
```

Output on next page

MatchID	Ground	Date	FName	LName
2755	Colombo	2008-08-24	Yuvraj	Singh
2755	Colombo	2008-08-24	MS	Dhoni
2755	Colombo	2008-08-24	Sanath	Jayasuriya
2755	Colombo	2008-08-24	Chaminda	Vaas
2755	Colombo	2008-08-24	Muthiah	Murlitharan
2689	Brisbane	2008-03-04	Sachin	Tendulkar
2689	Brisbane	2008-03-04	Yuvraj	Singh
2689	Brisbane	2008-03-04	MS	Dhoni
2689	Brisbane	2008-03-04	Adam	Gilchrist
2689	Brisbane	2008-03-04	Ricky	Ponting
2689	Brisbane	2008-03-04	Andrew	Symonds
2689	Brisbane	2008-03-04	Breet	Lee
2689	Brisbane	2008-03-04	Praveen	Kumar
2755	Colombo	2008-08-24	Praveen	Kumar
2689	Brisbane	2008-03-04	Harbhajan	Singh
2755	Colombo	2008-08-24	Harbhajan	Singh

2 [Inner Join with Filter]: Display Player full name, Country, NRuns, and Six for all batting performances where the player scored at least one six (Six > 0).

Solution:

SELECT P.FName, P.LName, P.Country, B.NRuns, B.Six FROM Player P INNER JOIN Batting B ON P.PlayerID = B.PID WHERE B.Six > 0;

FName	LName	Country	NRuns	Six
Sanath	Jayasuriya	Sri Lanka	60	2
Yuvraj	Singh	India	38	2
MS	Dhoni	India	36	1
Andrew	Symonds	Australia	42	1

3 [Left Outer Join]: Write a query using LEFT OUTER JOIN to list all players (PlayerID, FName, LName) along with their batting scores (MatchID, NRuns). Ensure players who have never batted in any match are also displayed with NULL values.

Solution:

SELECT P.PlayerID, P.FName, P.LName, B.MatchID, B.NRuns FROM Player P LEFT OUTER JOIN Batting B ON P.PlayerID = B.PID;

Output on next page

PlayerID	FName	LName	MatchID	NRuns
21001	Virender	Sehwag	NULL	NULL
23001	Yuvraj	Singh	2755	0
23001	Yuvraj	Singh	2689	38
24001	Andrew	Symonds	2689	42
25001	MS	Dhoni	2755	71
25001	MS	Dhoni	2689	36
27001	Praveen	Kumar	2689	7
27001	Praveen	Kumar	2755	2
27002	Ishant	Sharma	NULL	NULL
89001	Sachin	Tendulkar	2689	91
90001	Brain	Lara	NULL	NULL
90002	Anil	Kumble	NULL	NULL
91001	Sanath	Jayasuriya	2755	60
92001	Shane	Warne	NULL	NULL
92002	Muthiah	Murlitharan	2755	1
92003	Inzamam	UI-Huq	NULL	NULL
93002	Heath	Streak	NULL	NULL
93003	Gary	Kirsten	NULL	NULL
94002	Chaminda	Vaas	2755	17
94004	Stephen	Fleming	NULL	NULL
95001	Ricky	Ponting	2689	1
95002	Shaun	Pollock	NULL	NULL
95003	Jacques	Kallis	NULL	NULL

DBMS LAB 6

Parth Janardan Naukudkar_ECS_II_66_D6

4 [Right Outer Join]: Retrieve all bowling performance entries (MatchID, NOvers, Nwickets) along with player details (FName, LName, Country) using a RIGHT OUTER JOIN, showing all bowling entries even if player profiles are missing.

Solution:

```
SELECT B.MatchID, B.NOvers, B.NWickets,
    P.FName, P.LName, P.Country
FROM Player P
RIGHT OUTER JOIN Bowling B ON P.PlayerID = B.PID;
```

MatchID	NOvers	NWickets	FName	LName	Country
2689	10	1	Breet	Lee	Australia
2689	3	1	Andrew	Symonds	Australia
2689	3	0	Yuvraj	Singh	India
2755	9	1	Chaminda	Vaas	Sri Lanka
2755	10	1	Muthia	tharan	Sri Lanka
2755	4	0	Sanath	Jayasuriya	Sri Lanka
2755	10	2	Yuvraj	Singh	India
2755	10	3	Harbhajan	Singh	India
2689	10	1	Harbhajan	Singh	India

Muthiah

5 [Full Outer Join / Union]: Write an SQL query to perform a FULL OUTER JOIN (or simulate using LEFT JOIN UNION RIGHT JOIN) between Batting and Bowling tables on MatchID and PID to view all combined match participation records.

Solution:

```
SELECT B.MatchID, B.PID, B.NRuns, W.NOvers, W.NWickets FROM Batting
BLEFT JOIN Bowling W
   ON B.MatchID = W.MatchID
  AND B.PID = W.PID
```

```
UNION SELECT B.MatchID, B.PID, B.NRuns, W.NOvers, W.NWickets
FROM Batting B
RIGHT JOIN Bowling W
    ON B.MatchID = W.MatchID
   AND B.PID = W.PID;
```

Result Grid | Filter Rows:

MatchID	PID	NRuns	NOvers	NWickets
2755	23001	0	10	2
2755	25001	71	NULL	NULL
2755	91001	60	4	0
2755	94002	17	9	1
2755	92002	1	10	1
2689	89001	91	NULL	NULL
2689	23001	38	3	0
2689	25001	36	NULL	NULL
2689	99002	2	NULL	NULL
2689	95001	1	NULL	NULL
2689	24001	42	3	1
2689	99001	7	10	1
2689	27001	7	NULL	NULL
2755	27001	2	NULL	NULL
2689	98002	3	10	1
2755	98002	2	10	3

6 [3-Table Inner Join]: List Player.FName, Player.LName, ODIMatch.Ground, ODIMatch.Date, and Batting.NRuns for all matches played where 'India' was either Team1 or Team2.

Solution:

```
SELECT P.FName, P.LName, O.Ground, O.Date, B.NRuns FROM Player P
INNER JOIN Batting B ON P.PlayerID = B.PID
INNER JOIN ODIMatch O ON B.MatchID = O.MatchID
```

```
WHERE O.Team1 = 'India'
   OR O.Team2 = 'India';
```

FName	LName	Ground	Date	NRuns
Yuvraj	Singh	Colombo	2008-08-24	0
MS	Dhoni	Colombo	2008-08-24	71
Sanath	Jayasuriya	Colombo	2008-08-24	60
Chaminda	Vaas	Colombo	2008-08-24	17
Muthiah	Murlitharan	Colombo	2008-08-24	1
Sachin	Tendulkar	Brisbane	2008-03-04	91
Yuvraj	Singh	Brisbane	2008-03-04	38
MS	Dhoni	Brisbane	2008-03-04	36
Adam	Gilchrist	Brisbane	2008-03-04	2
Ricky	Ponting	Brisbane	2008-03-04	1
Andrew	Symonds	Brisbane	2008-03-04	42
Breet	Lee	Brisbane	2008-03-04	7
Praveen	Kumar	Brisbane	2008-03-04	7
Praveen	Kumar	Colombo	2008-08-24	2
Harbhajan	Singh	Brisbane	2008-03-04	3
Harbhajan	Singh	Colombo	2008-08-24	2

7 [4-Table Multi-Join]: Display player full name, Ground, Date, Batting.NRuns, and Bowling.Nwickets for all players who both BATTED and BOWLED in the exact same match.

Solution:

```
SELECT P.FName, P.LName, O.Ground, O.Date,
      B.NRuns, W.NWickets
FROM Player P INNER JOIN Batting B     ON P.PlayerID = B.PID
INNER JOIN Bowling W     ON B.MatchID = W.MatchID    AND B.PID = W.PID
INNER JOIN ODIMatch O     ON B.MatchID = O.MatchID;
```

FName	LName	Ground	Date	NRuns	NWickets
Breet	Lee	Brisbane	2008-03-04	7	1
Andrew	Symonds	Brisbane	2008-03-04	42	1
Yuvraj	Singh	Brisbane	2008-03-04	38	0
Chaminda	Vaas	Colombo	2008-08-24	17	1
Muthiah	Murlitharan	Colombo	2008-08-24	1	1
Sanath	Jayasuriya	Colombo	2008-08-24	60	0
Yuvraj	Singh	Colombo	2008-08-24	0	2
Harbhajan	Singh	Colombo	2008-08-24	2	3
Harbhajan	Singh	Brisbane	2008-03-04	3	1

8 [Self Join - Attribute Match]: Write a query using a SELF JOIN on the Player table to find pairs of players who were born in the same birth place (Bplace). Display Player 1 Name, Player 2 Name, and Bplace (exclude self-pairings and duplicate reversed pairs).

Solution:

```
SELECT P1.FName AS Player1_FName,
    P1.LName AS Player1_LName,  P2.FName AS Player2_FName,
    P2.LName AS Player2_LName,
    P1.Bplace
FROM Player P1 INNER JOIN Player P2  ON P1.Bplace = P2.Bplace  AND P1.PlayerID < P2.PlayerID WHERE P1.Bplace IS NOT NULL;
```

Player1_FName	Player1_LName	Player2_FName	Player2_LName	Bplace
Virender	Sehwag	Ishant	Sharma	Delhi
Gary	Kirsten	Jacques	Kallis	Cape Town
Gary	Kirsten	Herschelle	Gibbs	Cape Town
Jacques	Kallis	Herschelle	Gibbs	Cape Town

DBMS LAB 6

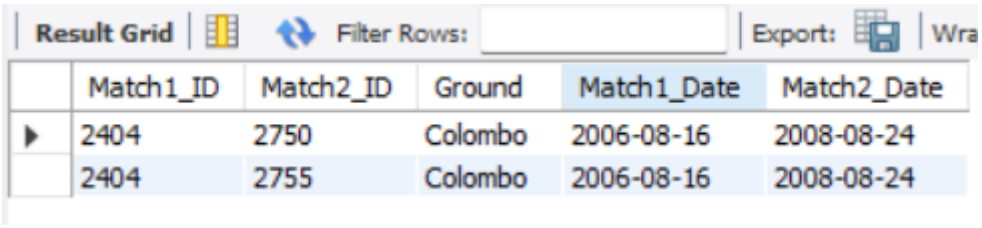
Parth Janardan Naukudkar_ECS_II_66_D6

9 [Self Join - Match Analysis]: Using a SELF JOIN on ODIMatch, find pairs of distinct matches that were played on the same Ground on different dates.

Solution:

```
SELECT M1.MatchID AS Match1_ID,
    M2.MatchID AS Match2_ID, M1.Ground, M1.Date AS Match1_Date,
M2.Date AS Match2_Date
FROM ODIMatch M1 INNER JOIN ODIMatch M2 ON M1.Ground = M2.Ground
AND M1.Date <> M2.Date AND M1.MatchID < M2.MatchID;
```

Output on next page



Result Grid | Filter Rows: | Export: | Wra

Match1_ID	Match2_ID	Ground	Match1_Date	Match2_Date
2404	2750	Colombo	2006-08-16	2008-08-24
2404	2755	Colombo	2006-08-16	2008-08-24

10 [Cross Join]: Generate a Cartesian product (CROSS JOIN) between players from 'India' and all matches in ODIMatch to display all hypothetical player-match combinations.

Solution:

```
SELECT P.PlayerID, P.FName, P.LName,
    O.MatchID, O.Ground, O.Date
FROM Player P CROSS JOIN ODIMatch O WHERE P.Country = 'India';
```

DBMS LAB 6

Parth Janardan Naukudkar_ECS_II_66_D6

PlayerID	FName	LName	MatchID	Ground	Date
406317	Sam	NULL	2324	Peshwar	2006-02-06
98002	Harbhajan	Singh	2324	Peshwar	2006-02-06
96003	Saurav	Ganguly	2324	Peshwar	2006-02-06
96001	Rahul	Dravid	2324	Peshwar	2006-02-06
90002	Anil	Kumble	2324	Peshwar	2006-02-06
89001	Sachin	Tendulkar	2324	Peshwar	2006-02-06
27002	Ishant	Sharma	2324	Peshwar	2006-02-06
27001	Praveen	Kumar	2324	Peshwar	2006-02-06
25001	MS	Dhoni	2324	Peshwar	2006-02-06
23001	Yuvraj	Singh	2324	Peshwar	2006-02-06
21001	Virender	Sehwag	2324	Peshwar	2006-02-06
406317	Sam	NULL	2327	Rawalpi...	2006-02-11
98002	Harbhajan	Singh	2327	Rawalpi...	2006-02-11
96003	Saurav	Ganguly	2327	Rawalpi...	2006-02-11
96001	Rahul	Dravid	2327	Rawalpi...	2006-02-11
90002	Anil	Kumble	2327	Rawalpi...	2006-02-11
89001	Sachin	Tendulkar	2327	Rawalpi...	2006-02-11
27002	Ishant	Sharma	2327	Rawalpi...	2006-02-11
27001	Praveen	Kumar	2327	Rawalpi...	2006-02-11
25001	MS	Dhoni	2327	Rawalpi...	2006-02-11

11 [Natural Join vs Explicit]: Write an SQL query joining Batting and ODIMatch using MatchID to display total runs scored by each player in matches played at 'Colombo' ground.

Solution:

SELECT B.PID, B.MatchID, B.NRuns, O.Ground FROM Batting B INNER JOIN ODIMatch O ON B.MatchID = O.MatchID WHERE O.Ground = 'Colombo';

PID	MatchID	NRuns	Ground
23001	2755	0	Colombo
25001	2755	71	Colombo
91001	2755	60	Colombo
94002	2755	17	Colombo
92002	2755	1	Colombo
27001	2755	2	Colombo
98002	2755	2	Colombo

DBMS LAB 6

Parth Janardan Naukudkar_ECS_II_66_D6

12 [Join with Aggregation]: Display PlayerID, FName, LName, and total career runs (SUM(NRuns)) across all matches by joining Player and Batting, sorted in descending order of total runs.

Solution:

```
SELECT P.PlayerID, P.FName, P.LName, COALESCE(SUM(B.NRuns), 0) AS TotalRuns
FROM Player P LEFT JOIN Batting B ON P.PlayerID = B.PID
GROUP BY P.PlayerID, P.FName, P.LName ORDER BY TotalRuns DESC;
```

Output on next page

PlayerID	FName	LName	TotalRuns
25001	MS	Dhoni	107
89001	Sachin	Tendulkar	91
91001	Sanath	Jayasuriya	60
24001	Andrew	Symonds	42
23001	Yuvraj	Singh	38
94002	Chaminda	Vaas	17
27001	Praveen		9
99001	Breet	Lee	7
98002	Harbhajan	Singh	5
99002	Adam	Gilchrist	2
92002	Muthiah	Murlitharan	1
95001	Ricky	Ponting	1
21001	Virender	Sehwag	0
27002	Ishant	Sharma	0
90001	Brain	Lara	0
90002	Anil	Kumble	0
92001	Shane	Warne	0
92003	Inzamam	UI-Huq	0
93002	Heath	Streak	0
93003	Gary	Kirsten	0
94004	Stephen	Fleming	0
95002	Shaun	Pollock	0
95003	Jacques	Kallis	0
96001	Rahul	Dravid	0

DBMS LAB 6

Parth Janardan Naukudkar_ECS_II_66_D6

13 [Left Join Anti-Join]: Write an SQL query using LEFT JOIN to identify all players (PlayerID, FName, LName, Country) who have NEVER recorded a bowling performance in the Bowling table.

Solution:

SELECT P.PlayerID, P.FName, P.LName, P.Country FROM Player P LEFT JOIN Bowling B ON P.PlayerID = B.PID WHERE B.PID IS NULL;

PlayerID	FName	LName	Country
21001	Virender	Sehwag	India
25001	MS	Dhoni	India
27001	Praveen	Kumar	India
27002	Ishant	Sharma	India
89001	Sachin	Tendulkar	India
90001	Brain	Lara	West Indies
90002	Anil	Kumble	India
92001	Shane	Warne	Australia
92003	Inzamam	UI-Huq	Pakistan
93002	Heath	Streak	Zimbabwe
93003	Gary	Kirsten	South Africa
94004	Stephen	Fleming	New Zealand
95001	Ricky	Ponting	Australia
95002	Shaun	Pollock	South Africa
95003	Jacques	Kallis	South Africa
96001	Rahul	Dravid	India
96002	Herschelle	Gibbs	South Africa
96003	Saurav	Ganguly	India
97004	Daniel	Vettori	New Zealand
98001	Shahid	Afridi	Pakistan
99002	Adam	Gilchrist	Australia
99003	Michale	Vaughan	England
300311	Michael	Clark	NULL